ARIAN RAHMAN ADITTA

Electronics & Communication Engineering Graduate | VLSI/EDA • Physical Design • ML/AI for EDA

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Academic Website | **LinkedIn** | **Google Scholar** | **GitHub**

RESEARCH INTERESTS

• VLSI & Electronic Design Automation • Physical Design Automation • ML/GNN for EDA • RTL-to-QoR Modeling

EDUCATION

B.Sc. Engineering in Electronics and Communication Engineering *[2022 - 2026]*

Khulna University of Engineering and Technology, Khulna, Bangladesh

CGPA: 3.71/4.00 | Class Rank: 5/60 **(Top 8.3%)**

Academic Distinction: Dean's Awards, **3rd and 4th** Year

Final four term GPAs: *3.89, 4.00, 4.00, 3.88*

Selected Coursework: VLSI Design & Nanotechnology, Analog Electronics, Microwave Engineering, Digital Communications, Antenna Engineering, Signals & Systems, Digital Signal Processing, Microprocessors & Microcomputers, Digital Electronics & Logic Circuits, Machine Learning

RESEARCH EXPERIENCE

GRACE: GNN-Guided Relational Decoding with Convex Shaping and Exact Repair for Constraint-Rich Floorplanning

Independent Research | 2026–Present | First-author manuscript submitted to ICEEICT 2027 | Ongoing

- Developed GRACE, where a GNN predicts block relations and soft-block shapes rather than final coordinates; deterministic relational decoding, convex shaping and exact repair keep the resulting floorplans hard-legal.
- Evaluated on 100 unseen FloorSet-Lite confirmation layouts, achieving 100/100 hard-legal outputs and outperforming the matched-time single-worker PARSAC CA-SA baseline on all 100 layouts.
- Designed leakage-free surrogate selection and controlled learned-start ablations; the learned initialization accounts for most of GRACE's wirelength advantage while exact repair combines constraint surgery, local moves and CP-SAT windows.

Source-Aware RTL-to-QoR Prediction for Early Area and Timing Estimation

Research Collaboration with Zakia Sultana, Ph.D. Candidate, MSCAD Laboratory, University of Arkansas | Ongoing | Manuscript in Preparation | GitHub

- Developed a source-aware RTL-to-QoR framework for mapped-area and standardized pre-layout timing prediction, separating source-derived structural representations from physical labels generated independently through Yosys/ABC and OpenSTA.
- Evaluated frozen Area and Timing systems prospectively on fresh paired cohorts; Area MAE improved from 8,382 to 3,015 μm² and Timing MAE from 0.167 to 0.136 ns, supported by robustness, runtime and reproducibility analyses.
- Extending the framework toward standardized RTL-to-power labeling and OpenROAD-based physical-design characterization, including floorplan, placement, CTS, routing and post-route timing, through reproducible open-source pilot pipelines.

Open-Source RFIC Design and Verification of a 28-GHz Two-Stage Class-A SiGe Power Amplifier with Machine-Learning-Assisted Optimization

Undergraduate Thesis, Department of ECE, KUET | Supervisor: Aysha Tabassum | 2026 | Ongoing

- Designed a **two-stage 28-GHz Class-A SiGe PA** using the **IHP SG13G2 130-nm SiGe BiCMOS Open PDK**, progressing from ideal-passive to PDK-aware and EM-aware realization through an open-source RFIC workflow.
- The final process-aware design achieved **17.4 dB gain at 28 GHz**; ML-assisted candidate screening was used as a supporting design tool, with a **first-author** thesis-derived manuscript submitted to **ICCIT 2026**.

27-GHz SiGe RF Front-End Design-Space Generation

Mentored Research with Sultan Muhammad Ikram, RF/mmWave MMIC Designer, Neural Semiconductor Limited | 2026–Present | GitHub

- Designed parameterized **common-emitter and cascode SiGe PA** topologies, systematically varying device sizing, bias conditions, RF chokes and input/output matching networks within the **IHP SG13G2** platform.
- Building a **private circuit parameter–performance dataset** using repeatable DC, small-signal, stability and large-signal characterization with Qucs-S, ngspice, Xyce and Python; extending the framework toward **common-emitter and cascode LNAs** for broader PA/LNA design-space analysis and ML-assisted RF design.

PUBLICATIONS & MANUSCRIPTS

Peer-Reviewed Publications

[1] **A. R. Aditta**, P. K. Das, U. R. Arshee, and A. B. M. Aowlad Hossain, "SenseStick: A Bangla-English Edge-AI Cane for Road-Safety and Biomedical Reliability Monitoring," *IEEE BECITHCON, 2026.* **[First Author]**

[2] P. K. Das, S. M. R. Islam, and **A. R. Aditta**, "LMatchNet-V: Physics-Seeded Learning for Vendor-Aware RF L-Section Matching with Adaptive Safe-Library Search," *IEEE MARSCON, 2026.*

[3] M. A. Jahin, S. Soudeep, **A. R. Aditta**, M. F. Mridha, N. Fahad, and M. J. Hossen, "Vision Transformers for End-to-End Quark-Gluon Jet Classification from Calorimeter Images," *Communications in Computer and Information Science*, vol. 2640, Springer, pp. 135–150, 2025. DOI: 10.1007/978-981-95-0988-1_10

Submitted Manuscripts

[4] **A. R. Aditta** and A. Tabassum, "Progressive PDK-to-EM Realization of a 28 GHz Two-Stage Class-A SiGe Power Amplifier in an Open-Source RFIC Flow," submitted to the *29th International Conference on Computer and Information Technology (ICCIT 2026)*. **[First Author]**

[5] **A. R. Aditta**, M. M. Zaman, M. S. U. Alam, and S. M. R. Islam, "GNN-Guided Relational Decoding with Convex Shaping and Exact Repair for Constraint-Rich Floorplanning," submitted to the *7th International Conference on Electrical Engineering and Information & Communication Technology (ICEEICT 2027)*. **[First Author]**

[6] "Source-Aware Evaluation of RTL Area and Timing Prediction with Prospective Confirmation." Manuscript in preparation.

TECHNICAL SKILLS

- **VLSI / EDA / RTL Tools:** Verilog, Yosys, OpenSTA
- **Programming & Scientific Computing:** Python, C, C++, MATLAB, NumPy, Pandas, Matplotlib
- **Machine Learning:** PyTorch, TensorFlow/Keras, scikit-learn, XGBoost, GNNs, CNNs, tree-based models, regression/classification, model evaluation
- **RFIC / Circuit / EM Tools:** Qucs-S, ngspice, Xyce, KLayout, openEMS, CST Studio Suite, LTspice, Cadence Virtuoso, IHP SG13G2 Open PDK
- **Embedded Systems:** Raspberry Pi, ESP32, Arduino, OpenCV
- **Research Tools:** Git, LaTeX, Zotero

HONORS & AWARDS

Dean's Award, 3rd & 4th Year, Khulna University of Engineering & Technology
- Awarded for outstanding academic performance in consecutive senior academic years.

Technical Scholarship, Khulna University of Engineering and Technology (KUET)
- Awarded for academic performance during the first three undergraduate years.

ACADEMIC & RESEARCH LEADERSHIP

President, KUET Research Society, ECE *[2025-2026]*
- Organized a department-wide pre-mid thesis competition involving 3rd- and 4th-year ECE students to promote research presentation, peer learning and early thesis engagement.
- Coordinated online graduate-study and research guidance sessions with KUET alumni pursuing M.S./Ph.D. degrees abroad, helping current students understand research preparation, graduate applications, and academic career pathways.

General Secretary, KUET Debating Society *[2023-2026]*
- Led organizational activities and competitive debating initiatives, developing extensive experience in communication, team leadership, public speaking, team coordination and event management.

RELEVANT PROFESSIONAL EXPERIENCE

Campus Ambassador, Ulkasemi *[Aug 2024 – Sep 2025]*
- Supported KUET–Ulkasemi semiconductor-industry engagement and student outreach; helped organize a five-day career and management training program for final-year students.

Data Analyst Intern, WEGRO *[Mar 2025 – Jul 2025]*
- Analyzed business data, created decision-oriented visualizations, and contributed to data-driven process improvement in an agri-fintech startup.